

CURRICULUM VITAE

Luqman Munir, M.B.B.S.

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PROFESSIONAL SUMMARY

Medical student researcher specializing in ophthalmology AI applications and epidemiological analysis. Currently conducting retinal imaging research at Johns Hopkins Wilmer Eye Institute with expertise in CycleGAN development, TriNetX database analysis, and systematic review methodologies. Published 21 peer-reviewed articles with focus on population health disparities and evidence-based clinical interventions.

EDUCATION AND TRAINING

Medical Education

2021 - present M.B.B.S., King Edward Medical University, Lahore, Punjab, Pakistan

Pre-Medical Education

2019 - 2021 F.Sc. Pre-Medical, Punjab College of Science, Islamabad, Federal Capital Territory, Pakistan

USMLE Examination Scores

Step 1 Pass

PROFESSIONAL EXPERIENCE

Research Experience

Research Intern, Paulus Lab, Retina Department

07/2025 - present

Johns Hopkins Wilmer Eye Institute, Baltimore, MD

- Machine learning models on studying the disease progression in Diabetic Retinopathy
- A machine learning model and an online calculator on retinal detachment surgery risk.
- Large database based retrospective analysis of a multitude of retinal pathologies and interventions
- Conducted real-world cost analysis of Eylea HD
- Developed CycleGAN prototype achieving 85% accuracy in fundus→OCT cross-modal imaging translation
- Co-authoring a review on artificial intelligence applications in retinal disease
- Contributing to age-related macular degeneration (AMD) research
- Launching a new AI-focused retinal detachment research project
- Applying large language models to support research study design and literature synthesis
- Investigating steroid versus anti-VEGF therapy outcomes
- Studied the association between methotrexate use and recurrent retinal detachment
- Contributing to a ChatGPT-based health AI project in ophthalmology
- Built an automated EHR data extraction pipeline for structured retinal detachment surgical outcomes research

- Developed a deep learning image segmentation pipeline for retinal detachment biomarker analysis on OCT scans

Visiting Student, Wilmer Glaucoma ML Lab, Dr. Yohanan
Johns Hopkins Wilmer Eye Institute, Baltimore, MD

09/2025 - present

- Studying the effects of Mitomycin C on shunt surgeries
- Helping in the development of a clinical decision support tool
- Comparing subtenon corticosteroid injection versus trabeculectomy outcomes
- Developing an AI-based target intraocular pressure (IOP) prediction tool
- Analyzing outcomes of glaucoma drainage device tube revision surgery

Visiting Student, Wilmer Eye Research Institute, Dr. Fasika Woreta
Johns Hopkins Wilmer Eye Institute, Baltimore, MD

09/2025 - present

- Studying open globe injuries and risk of endophthalmitis
- Studying the worldwide cataract curriculum
- Contributing to a Delphi consensus study
- Studying postoperative day 1 (POD1) intraocular pressure spikes
- Built Epic EHR data extraction and analysis pipelines (SQL, Python) to support open globe injury and cataract curriculum research

Research Intern, Department of Genetics
Lahore University of Management Sciences, Lahore, Pakistan

06/2024 - 8/2024

- Conducted genomic data analysis using R and bioinformatics tools
- Contributed to population genetics research on South Asian disease susceptibility variants

CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT

Johns Hopkins University

- Introduction to Genomic Technologies (May 2025)
- Introduction to Systematic Review and Meta-Analysis (May 2025)
- Design and Interpretation of Clinical Trials (May 2025)

Harvard University (EdX)

- Statistics and R Programming (May 2025)

Neuromatch Academy

- Computational Neuroscience (August 2023)

National Institute of Health

- Introduction to the Principles and Practice of Clinical Research (July 2023)

LANGUAGES

English: Fluent (Professional working proficiency)

Urdu: Native speaker

Punjabi: Native speaker

Arabic: Basic conversational

RESEARCH ACTIVITIES

Current Research Projects

Artificial Intelligence Projects

- CycleGAN Fundus→OCT Prototype Development (Johns Hopkins Wilmer Eye Institute)
- Longitudinal AI Model for Retinal Diagnosis and Disease Progression (Johns Hopkins Wilmer Eye Institute)

Clinical Research Projects

- Real World Cost Analysis of Eylea HD®: A Retrospective Analysis of Real-World Data
- Micropulse Laser vs. Anti-VEGF Injections for Diabetic Macular Edema (TriNetX analysis)
- Systemic Hypoxia and Diabetic Retinopathy Progression (TriNetX analysis)
- eIRB Review: Retinal Outcome Study (Johns Hopkins Wilmer Eye Institute)

Software and Data Infrastructure Projects

- Paulus Lab Research Dashboard: full-stack project, task, conference, and resource management platform (Next.js, PostgreSQL) built and deployed for lab-wide use, including a mobile-optimized interface
- Woreta Lab Research Dashboard: adaptation of the lab dashboard platform for the Wilmer Eye Research Institute (Dr. Fasika Woreta), with role-based access control
- Retinal Detachment (RRD) EHR Extraction Pipeline: automated structured extraction of surgical and clinical data from Epic Clarity/EDW combining SQL and LLM-based clinical note parsing
- OCT Image Segmentation Pipeline: deep learning (SegFormer-B4) segmentation model with active learning (MONAI Label) for automated retinal detachment biomarker identification on OCT scans
- Ophthalmology Faculty Outreach Agent: multi-agent AI system (CrewAI) automating faculty discovery, profiling, and personalized outreach across ACGME ophthalmology programs

Research Skills and Technical Expertise

Database Analysis and Statistical Methods

- TriNetX query design and implementation
- SQL/PMAP data extraction and manipulation
- CDC WONDER database analysis
- Global Burden of Disease (GBD) database studies

- R programming for statistical analysis and data visualization
- Advanced biostatistical methods including survival analysis, time-series analysis

Artificial Intelligence and Machine Learning

- CycleGAN implementation and optimization
- Python programming for deep learning applications
- Computer vision applications in medical imaging
- Longitudinal data modeling for disease progression
- Semantic segmentation and active learning workflows for medical imaging (SegFormer, MONAI Label)
- Full-stack web application development (Next.js, PostgreSQL, Prisma) for research infrastructure
- Multi-agent LLM systems (CrewAI) for research automation

Research Methodology and Regulatory

- eIRB proposal writing and submission
- Grant proposal development and writing
- Systematic review and meta-analysis methodology
- Clinical trial design and interpretation
- Research ethics and regulatory compliance

PRESENTATIONS

Major Conference Presentations Summary

2024 Presentations

- Congress of Neurological Surgeons Annual Meeting: 14 posters
- American College of Gastroenterology Annual Scientific Meeting: 5 oral presentations, 5 posters
- American Society of Nephrology Kidney Week: 4 oral presentations, 6 posters

2025 Presentations

- American Surgical Congress Session: 5 oral presentations
- American Academy of Neurology: 3 oral presentations, 2 posters
- International Congress of Parkinson's Disease and Movement Disorders: 2 posters
- World Stroke Congress: 4 posters
- SuriCon: 1 poster

Total Conference Impact: 17 oral presentations, 39 poster presentations across major international medical conferences

ORGANIZATIONAL ACTIVITIES

Leadership Positions

Co-Founder and Head, NeuralKey 04/2024-present
Neuroscience-based interest group, King Edward Medical University, Lahore

- Founded national neuroscience platform with 200+ members across Pakistan
- Established research mentorship program training 25+ undergraduate researchers
- Secured \$5,000 funding for symposium hosting 150+ attendees

Research Head, KemCardio 07/2024-present

Cardiology Interest Group, King Edward Medical University, Lahore

- Oversee 12 active research projects
- Implemented quality review process reducing manuscript revision cycles by 35%

Research Division, KemVision 09/2024-present

Ophthalmology Interest Group, King Edward Medical University, Lahore

Professional Memberships

2025-present Member in Training, Association for Research in Vision and Ophthalmology (ARVO)

EDITORIAL BOARD APPOINTMENTS

2025-present Student Council Member, Journal of Society of Prevention, Advocacy and Research

EDUCATIONAL ACTIVITIES

Teaching and Mentoring

- Research Methodology Workshop Leader: "Systematic Review and Meta-Analysis" (NeuralKey)
- Guest Lecturer: "CDC Wonder Database Analysis" (NeuralKey, King Edward Medical University)
- Pre-Medical Course Instructor: Basic Science Fundamentals (MedAngle, Karachi)

PUBLICATIONS

Primary Studies [Original Clinical Research]

1. Xie X, **Munir L**, Paulus YM. Retinal Laser Therapy Mechanisms, Innovations, and Clinical Applications. *Photonics*. 2025; 12(11):1043. <https://doi.org/10.3390/photonics12111043>
2. Qureshi MA, Haseeb S, Goel M, **Munir L**, et al. Inpatient outcomes of transcatheter aortic valve replacement in patients with underlying hypercoagulable states. *J Soc Cardiovasc Angiogr Interv*. 2025;4(5). doi:10.1016/j.jscai.2025.103332
3. **Munir L**, et al. Establishment of a centralized civil registration and vital statistics (CRVS) database in Pakistan: a healthcare perspective. *BMC Health Serv Res*. 2024. [In Press]
4. **Munir L**, et al. Building a stronger backbone: 3D printing's role in treating spinal cord conditions. *J Spinal Cord Med*. 2024. [In Press]

Review Studies [Systematic Reviews and Meta-Analyses]

4. **Munir L**, et al. Evaluating risk factors of embolism in patients with cardiac myxoma: A systematic review and meta-analysis. *Am Heart J Plus*. 2025;100559. doi:10.1016/j.ahjo.2025.100559
5. **Munir L**, et al. Assessing the impact of screen time on the motor development of children: A systematic review. *Child Dev Perspect*. 2024. [In Press]
6. **Munir L**, et al. Efficacy and safety of LipiFlow® in the treatment of dry eye disease due to Meibomian gland dysfunction: an updated meta-analysis of randomized controlled trials. *Ocul Surf*. 2024. [In Press]
7. Qureshi AA, **Munir L**, Alvi Z. Etidronate for arterial calcification due to deficiency of CD73 (ACDC): A step in the right direction. *J Rare Dis*. 2024;3(1):35. doi:10.1007/s44162-024-00059-6
8. Shahzil M, Husnain A, Nadeem A, **Munir L**, Qureshi A, Shaukat M. Efficacy and safety of prophylactic transarterial embolization combined with endoscopy in high-risk peptic ulcer disease patients: A meta-analysis. *J Vasc Interv Radiol*. 2025;36(3):S92-S93. doi:10.1016/j.jvir.2024.12.249
9. Shahzil M, Kashif TB, Jamil Z, **Munir L**, et al. Assessing the effectiveness of texture and color enhancement imaging versus white-light endoscopy in detecting gastrointestinal lesions: A systematic review and meta-analysis. *DEN Open*. 2026;6(1):e70128. doi:10.1002/deo2.70128

Epidemiological Studies [CDC and GBD Database Analysis]

10. **Munir L**, Chaudhary AJ, Ammar ur Rahman M, Khalid A, Ijaz A. Trend analysis of kidney stone-related mortality, 1999-2020. *J Am Soc Nephrol*. 2024;35(10). doi:10.1681/ASN.0000000000000480
11. Singh H, Qureshi M, **Munir L**, Qureshi A. Tuberculosis mortality rates in the united states from 1999-2020: An area of concern. *Crit Care Med*. 2025;53(1).
12. Shahid I, Ur Rahman MA, Shahid F, Fatima M, Raza A, **Munir L**. Trends in epilepsy related mortality from 1999–2020: An analysis of the CDC wonder database. *Neurology*. 2025;104(7):1601. doi:10.1212/WNL.0000000000208635
13. Shahid F, Raza A, Ur Rahman MA, **Munir L**. Evolving patterns of myasthenia gravis and other myoneural disorders related mortality from 1999–2020: An analysis of CDC wonder database. *Neurology*. 2025;104(7):3127. doi:10.1212/WNL.0000000000210813